SFU SIMON FRASER UNIVERSITY Visual & Interactive Computing Concentration ***Fall 2024 GRADUATION PLANNER***	***COMPUTING SCIENCE B.SC. MAJOR —***

Name: ______________________________ Student ID: ________________________ Date: ________________

Visual & Interactive Computing covers computer graphics, human-computer

Year 1

☐ CMPT 120 Programming 1
☐ MACM 101 Discrete Math 1
☐ MATH 151 Calculus 1
OR ☐ MATH 150 Calculus 1 with Review
☐ CMPT 105W CS Writing I (or in 2nd term)
☐ WQB Breadth: _____________

☐ CMPT 125 Programming 2
☐ MATH 152 Calculus 2
☐ MATH 240 Algebra I: Linear Algebra
OR ☐ MATH 232 Applied Linear Algebra
☐ WQB Breadth: _____________
☐ General Elective: ____________

Year 2

☐ CMPT 225 Data Structures and Programming
☐ CMPT 201 Introduction to Computer Science and Programming for Engineers / CS core
☐ CMPT 295 Intro to Computer Systems
☐ CMPT 210 Probability and Computing
OR ☐ MACM 201 Discrete Mathematics II
☐ WQB Breadth: _____________

☐ CMPT 276 Software Engineering
☐ STAT 271 Probability and Statistics for CS
OR ☐ STAT 270 Introduction to Probability & Statistics
☐ WQB Breadth: _____________
☐ WQB Breadth: _____________
☐ General Elective: ____________

Year 3

☐ CMPT 361 Introduction to Computer Graphics
☐ CMPT 307 Data Structures and Algorithms
☐ MACM 316 Numerical Analysis
☐ CMPT 376W CS Writing II
☐ General Elective: _______________

☐ CMPT 300 Operating Systems
☐ CMPT 363 User Interface Design
☐ CMPT 365 Multimedia Systems
☐ General Elective: _______________
☐ General Elective: _______________

Year 4

☐ CMPT 461 Advanced Computer Graphics*
☐ CMPT 354 Database Systems
☐ CMPT 379 Compilers
☐ UD General Elective: ___________
☐ General Elective: ____________

☐ CMPT 466 Computer Animation*
☐ CMPT 467 Survey of Image Synthesis Techniques*
☐ CMPT 469 Sensor Technology and Its Applications*
☐ UD General Elective: ___________
☐ General Elective: ____________

* Or any other 400-level AI Elective: CMPT 410 Machine Learning, CMPT 411 Knowledge Representation, CMPT 412 Computational Vision, CMPT 413 Computational Linguistics, CMPT 414 Model-Based Computer Vision, CMPT 417 Intelligent Systems, CMPT 419 Special Topics in Artificial Intelligence, CMPT 420 Deep Learning.

Other recommended general electives: CMPT 320 Social Implications, STAT 350 Linear Models In Applied Statistics, STAT 440 Learning from Big Data (requires 90 units & STAT 350), CMPT 456 Information Retrieval & Web Search.

WQB Breadth Requirements

- 6 units of Breadth Social (B-SOC)
- 6 units of Breadth Humanities (B-HUM)
- 3 units of Breadth Science (B-SCI)

Refer to http://www.sfu.ca/ugcr/for_students/wqb_requirements/breadth.html for courses that fulfill these requirements.

This Concentration Planning Form contains a recommended course plan for Computing Science major BSc students to obtain a concentration designation, along with course suggestions to optimize the knowledge and skills upon completion of this concentration, while distributing the difficulty of the course load per term. Other course plans may be possible. This form is not a substitute for the official degree regulations found at www.sfu.ca/students/calendar.html. If there is a question of interpretation or discrepancy, the University Calendar always takes precedence. For assistance or queries on possible substitutions, ask a FAS advisor to help. The student is ultimately responsible for ensuring that they have met their degree requirements.

CO-OPERATIVE EDUCATION Combines work experience with academic studies—all students are encouraged to apply once they have completed 30 units. Co-op does not count towards academic credits. Co-op is not mandatory; however, three work terms must be successfully completed in order to obtain an undergraduate degree with a co-op designation. For more information about Co-op, please see: https://www.sfu.ca/coop/programs/cmpt/prospective.html.

CMPT 415/416 SPECIAL RESEARCH PROJECTS are courses that may be used for upper division credit. See: https://www.sfu.ca/computing/current-students/undergraduate-students/research.html

FACULTY OF APPLIED SCIENCE RESIDENCY REQUIREMENTS At least two thirds of the total Upper Division (UD) units in the program must have been completed at Simon Fraser University. Please refer to the current SFU calendar for details.

CONTINUATION REQUIREMENTS Students who do not maintain at least a 2.40 CGPA, will be placed on probation by the School of Computing Science. Courses available to probationary students may be limited. Each term, these students must consult an advisor prior to enrollment and must achieve either a term 2.40 GPA or an improved CGPA. Students who fail to do so may be removed from the program.

ADVISING View drop-in advising times here https://booking.cs.sfu.ca/adbooking/calendar.cgi or email asadvise@sfu.ca. Please bring a copy of your advising transcript (download at go.sfu.ca) with you to the advising session.

Visual & Interactive Computing Concentration Planner
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